YEAR 8 MATHS · CHAPTER 6 · MIND MAP

Ratios and rates

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1 Simplifying ratios

Divide both parts by the HCF. Use the same units first.

$12 : 18 = 2 : 3$

50 cm : 2 m $= 50 : 200 = 1 : 4$

$0.5 : 2 = 1 : 4$

Three parts: $6 : 9 : 15 = 2 : 3 : 5$

Watch out: convert to the same units before you simplify.

2 Solving ratio problems

Add the parts, find one part, then multiply.

$60 in the ratio 2 : 3: 5 parts, 1 part = $12

So **$24** and **$36**

In the ratio 3 : 5 with the first part 12: the second is 20

$90 in the ratio 1 : 2 : 3: 6 parts, 1 part = $15, so **$15**, **$30** and **$45**

3 Scale drawings

A scale of 1 : 100 means 1 cm on the drawing is 100 cm in real life.

1 : 100, 4.5 cm on the plan: $4.5 \times 100 = 450$ cm, or 4.5 m

1 : 50 000, 3 cm on the map: 150 000 cm, or 1.5 km

Finding the scale: a 6 m wall drawn 3 cm long is 3 cm : 600 cm = **1 : 200**

Watch out: use the same units on both sides of the scale.

4 Introducing rates

A rate compares two different units. Read "per" as "for each".

150 km in 3 hours: **50 km/h**

$12 for 4 kg: **$3/kg**

Best buy: 2 L for $5 is $2.50/L; 3 L for $6.60 is $2.20/L, so the **3 L** is better value

5 Solving rate problems

Multiply the rate by the time, or divide to find the time.

8 L/min for 30 min: $8 \times 30 = 240$ L

200 L at 8 L/min: $200 \div 8 = 25$ min

Typing 45 words a minute for 20 minutes: $45 \times 20 = 900$ words

Check the units match before you multiply or divide.

6 Speed

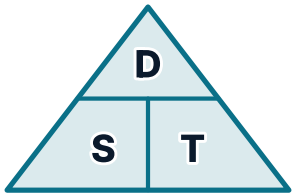
$S = \frac{D}{T}, D = S \times T, T = \frac{D}{S}$

240 km at 80 km/h: $T = 3$ h

90 km in 1.5 h: $S = 60$ km/h

45 minutes at 80 km/h: $D = 80 \times 0.75 = 60$ km

72 km/h in m/s: $72 \div 3.6 = 20$ m/s



Watch out: 45 minutes is 0.75 hours, not 0.45.

7 The unitary method

Find the value of one, then multiply to get what you need.

5 pens cost $8: 1 pen is $1.60, so 12 pens are $19.20

In a ratio, 3 parts = 45: 1 part is 15, so 7 parts are 105

A 250 g pack costs $4.50: 100 g costs $4.50 \div 2.5 = \$1.80$

Finding the value of one works for any amount.